

Accelsius 2026.4.17 Josh Claman

A: 28M @ \$135M

B: Strategics \$65M @660 JCI @35M; Legrand \$20M

- Legrand – 30 accounts going to be in thermal
- JCI – Enough to hit there five year numbers
- Grey and white space will combine

SLM: Small Language models – on-prem with corporate

- 2027 on will be inference (latency issues)

Hyperscalers:

- 2-3 now and will cool – working with hyperscalers now
- Vertiv – go through channel and white label with them – channel will be able to sell it
- Not loved by clients (Vertiv)

TwoPhase:

- Will be dominant player
- Eco-labs and Vertiv are working on it – 2Phase is hard – its complicated to simulate and mechanically engineer it, pressure water
- Vertiv is 2 years behind us
- NVIDIA won't talk about 2 Phase for another 18 months
- Innovator dilemma – people are slow to transition at this point (need the capital return)
- AMD – Helius racks with AMD GPUs
- Delta Taiwan – working with them to work with phase 1 and phase 2
- Adoption is the biggest risk

Semi-analysis:

- Once over 1500W then have to go to 2 phase – next chip is going to be 2200 they are going to be needed for 2 phase
- “better now – required later”
- Capex is the same – and 35% lower on the opex
- 10 MW doesn't matter energy savings – 500MW or 1GW – the energy savings is so massive (would power a city)
- Cooling mkt – 1.1B; 2.6B; 3B; 7.3B; 13.2B; 18.9B; 30.3B 2024-2030
- Accelsius Revenue: 0; 0; 90; 171; 290; 479; 719M; 2024-2030

- 2 Phase: 0;26;150; 510; 1.3B; 3.8B; 9.1B 2024-2030
- A Mkt share: 0; 0; 60%; 34%; 22%; 13%; 8% 2024-2030
- 2027/2028 offer can't refuse (10+ buyers)
- Prettiest girl at the dance only one left
- Revenue hasn't hit – need some revenue
- 10-12x sales comps
- Don't want to run a public company
- Probable scenario – 1,2,3B some offer will be so big
- Fintech and High frequency traders will be the highest adoption players

Series C:

- Don't need cash for 18 months
- Only if w/c grows from massive growth –
- Direct to another transaction
- Doesn't like INV

Competitor:

- Not worried about Zukacore – Mitsubishi investing – if they were better would help them out
- Vertiv – every tree
- Boyd – Eaton bought them – does have some 2 phase expertise (mostly aerospace)

AI Notes:

Accelsius Investor Onsite

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Company Overview & Strategic Positioning

- Accelsius celebrating 4th birthday in July - young for hard tech sector
- Third startup for CEO Josh, explicit strategy to avoid VCs in A round

- Raised ~\$28M in A round at \$135M post-money from high net worth/family office investors
- B round focused on strategic investors for credibility and market reach

Strategic Partnerships

- Johnson Controls: \$35M investment
 - 120 years thermal management expertise, no white space data center presence
 - Multi-billion dollar gray space data center business
 - Joint comprehensive architecture development planned
- LeGrand: ~\$7B revenue, \$40B market cap
 - First minority investment for them, very acquisitive company
 - 30 joint accounts currently in pipeline
 - Co-development of liquid cooled busways and prefab compute pods
- Vertiv: OEM relationship for appliance-based model
 - 18-month new product introduction process slowing deployment

Technology & Market Dynamics

- Two-phase cooling universally acknowledged as future standard
- Current forcing function: GPU power moving from 1500W to 2200W+ per socket
- Nvidia privately admits single-phase water architecture is flawed but won't publicly endorse two-phase for 18+ months
 - Protecting relationships with Vertiv/Schneider investments
 - Flow rates causing nickel leaching, server damage, maintenance issues
- AMD more aggressive on two-phase adoption - Accelsius will be reference design partner

Competitive Landscape & Adoption Challenges

- Total liquid cooling market: \$1B (2024) → \$30B projected
- Two-phase market: \$25M (2025) → \$9B projected by 2030

- Major competitors: Vertiv (2 years behind), Boyd/Eaton acquisition concern
- Zudicore struggling with product quality, failed Hudson River tests
- Conservative data center culture vs. revolutionary technology changes creating adoption friction

Product Portfolio & Manufacturing

- MR-250: Current shipping product, 250kW capacity, being recertified to 350kW
- IR-150: Integrated rack solution for Vertiv, completely plug-and-play
- MR-2K: 2MW CDU in development for hyperscaler requirements
- Manufacturing: Current facility supports 18-month revenue goals, 25% internal/75% CM strategy planned

Financial Projections & Market Opportunity

- 2025 target: \$90M revenue in \$150M total market
- Conservative market share assumptions with declining share as market grows
- 10-12x revenue multiples observed in recent acquisitions
- Multiple acquisition inquiries received, not ready to sell yet
- Probable exit scenario 2027-2028 timeframe

Customer Pipeline & Validation

- Two hyperscaler contracts active, third in arrangement phase
- Dark NX: \$55M contract, multiple sites, construction already started
- Hudson River Trading and other fintech early adopters expected this year
- Dell invited Accelsius to executive briefing center
- AMD partnership launching June with consigned Helios rack at Meta

Key Technical Advantages

- Superior heat removal with phase change physics
- 35% OPEX advantage through warmer water loops
- Non-conductive fluid prevents equipment damage from leaks
- Infrastructure compatibility - plug-and-play with existing water systems

- Future-proof for increasing chip power densities

Regulatory & Market Tailwinds

- Senate Commerce Committee interest in data center efficiency mandates
- Community protests stopping ~\$200B in large data center projects
- Trend toward smaller, distributed inference pods vs. massive LLM facilities
- Insurance premiums rising due to thermal architecture failures

Next Steps & Immediate Priorities

- Board reconstitution by end of year with industry heavy hitters
- Series C not planned - direct to exit transaction preferred
- Universality testing with Delta to prove single SKU for water/two-phase
- Senate Commerce Committee speaking engagement upcoming
- Continued hyperscaler POC progression to production deployments

Transcript:

Meeting Title: Accelsius Investor Onsite

Date: Apr 17

Meeting participants: Mike, Rtonkel, Chrismaxtonegraham, Abrosey61, Josh Claman, Brynkahrl, Craig, Ddiaz, Demos&on-sites, Einstein, Gavin, Pcrowley, Rgreg32, Stevenchoran, Ted Hoffman

Transcript:

Speaker A: What our. What our hopes and dreams are. Fair enough.

Speaker B: Sounds good.

Speaker A: Love to go around with introductions. And Diego's joining here as well. Many of you probably know Diego from Adventure. Yeah, sure.

Speaker B: Stephen Schorn, worked on Wall street for 25 years, retired in 18, and now I'm a personal investor.

Speaker A: Carl. I'm still working in a business approaching maybe retirement here this year. And you know, personal investment. Good Adventure. You know, I've tried retiring twice. I failed miserably.

Speaker B: You got to come to Miami. It's a lot easier.

Speaker A: Probably true.

Speaker B: My career was New York.

Speaker A: My career was in manufacturing, so I started in technology and went to automotive. Oh, fantastic. So pretty relevant, actually. Different field, but pretty relevant to what we're doing as well. I'm excited to see it. Okay.

Speaker C: Hi, I'm Mike, based in Dallas. Invest through a family office.

Speaker D: Chris Graham, returning holder of this fine company. Spent about 15 years on Wall Street. Have my own investment firm, Vale Management, based in Dallas.

Speaker A: Ted Hoffman, also an investor of a Celsius and yeah, happy to be back. Great, great. Is that Eric? Maybe Eric joined us. I don't know. Maybe. I think he's here. I think he's here. He is?

Speaker E: Yeah, he is.

Speaker C: Let me go.

Speaker A: Oh, he's here. Oh, okay.

Speaker F: Rock Talkle representing Clear Street, New York based investment bank.

Speaker A: Give him a shot. Gavin Ritchie, Clear street client. Run an investment business in Dallas.

Speaker F: Craig Samuels, been investing capital for 30 years.

Speaker E: Greg Reyes, retired CEO, manage my own family office.

Speaker G: Alan Rosenthal, capital markets advisory firm here out of Boston. Was on Wall street for a long time and sorry to the clearest few folks.

Speaker A: We placed the \$40 million

Speaker G: at Titan. So I'll put that with that.

Speaker A: I'm Matt Sita.

Speaker F: I'm Allen's the new coo.

Speaker A: Nice to meet you.

Speaker E: Perfect timing.

Speaker B: I'm Diego.

Speaker F: I'm the chief of staff here at

Speaker A: Excelsis and previously at Inventure. Just pull up some slides here. It's clear.

Speaker F: Straight to affirm that pulled its ipo.

Speaker A: Yeah.

Speaker F: Couple weeks ago. Yeah, months ago at this point.

Speaker A: Yes. It's a great place though, unfortunately.

Speaker F: Are you guys rolling out a new trading platform soon? Yeah, yeah. More detail on that probably imminently hitting the wire and wait for that.

Speaker A: Right. So we do maybe a show of hands. How many people are sort of ingrained in the data center sector in some way? So few. So I'll try to minimize acronyms and data. Data center speak. But I think there's a lot we can convey without. Without getting into too much jargon. We have been around for. So we're going to celebrate our fourth birthday in July. So we're still a relatively young company. That's pretty young in hard tech. You know, in SaaS or software four years is pretty adolescence. But in hard tech it's a little younger because you have to go through multiple rounds of prototyping, builds, et cetera, certification to get products out. All those sort of roadblocks. But we feel like we've made really good progress. We were seeded by Inventures. Many of you know. We then did an A round which was totally direct to a lot of you were involved, mostly high net worth, family office type investors. That was my explicit strategy. We were approached by a lot of VCs to do to enter the A round. I didn't want VCs in the a round. They typically try to fit you into a box and say well we think your valuation should be this and we want to own 23% of you and this and that. I didn't like any of that talk. And this is my third startup and there's sort of a learning I've gained over time. The B round we raised about 20, I think total 28 million or so in the A round at a post money of about 135. The B round I explicitly wanted to bring in not institutional investors but strategics that could give us credibility, give us market reach, go to market with us, even do co development with us. And so we started courting a few. Johnson Controls was one of the first. They put in \$35 million. Johnson Controls has 120

years in thermal management, a lot of expertise in two phase physics particularly. But more importantly they have no presence in the white space in the data center. That's where all the server racks are, that's where all the it is. They are huge in the gray space of the data center. So you know, chiller infrastructure, pumping infrastructure, that sort of thing. They're very busy. They have a multi billion dollar business in the data center sector. They think they're too small in the data center sector. So I got the two advantages of a lot of presence that could drag us into a lot of accounts, give us credibility, give us thermal credibility. But they were still hungry. So I got kind of both sides of that. The idea with Johnson Controls is that companies you can imagine being a data center business builder or an operator today and you want to build a big Data center, say 100 megawatts to a gigawatt and you see a new gigawatt data center announced every Tuesday in this sector we can come back to that theme as well, and what's real and what's not. But they have to do some sort of liquid cooling. Their rack densities per server rack have gone from four and a half to seven to eight kilowatts suddenly to 100 kilowatts, 200 kilowatts. And now we're talking about half a meg or megawatt racks. So how do you cool that? No one really knows. How do you power that? No one really knows. Right. I'm being a little bit glib, but I'll talk about what I mean by nobody knows as we go through this. But the idea is that you have these operators from what has been a really conservative sector. Data centers really haven't changed for 40 years. You know, they, they refine air conditioning. They contain hot aisles slightly more effectively. Their rack densities went from say, 2 kilowatts of rack to five and a half, 6 kilowatts of rack over 30 years. And suddenly in right now, you have to go to 100 kilowatt racks or 150 kilowatt racks. And your data center that you thought was big at, you know, 5 megawatts is now kind of an inference pod. It's, it's nothing. Right. How do you build a 1 gigawatt data center? So they're getting a lot of people coming to them like Excelsior saying we're the best in rack cooling or we're the best at power conditioning. Hey, Eric. So what we wanted to do was join forces with Johnson Controls and say, don't worry about being pecked to death by people like Excelsius or other cooling companies. We'll develop a comprehensive sort of shift to atmosphere architecture with data behind it. So we'll have testing behind it that says we'll do the chiller infrastructure that's, that's keyed into the IT cooling infrastructure. And we'll be able to tell you what your efficiency is in that. It's called PUE or TUE in that data center. Right. So we'll be able to model that for you so it makes their life easier. They don't have to climb up so many individual learning curves. And there's just so much noise and claims in the market about what they do. Right. So I keep coming back to the theme, very conservative market. These people are more into job protection. No one gets fired for buying IBM. That was still true five years ago in the data center market. I mean, you went through this as you created new business, but it's still true today. You have Two, oil and water. Very conservative nature of data centers mixed with the most revolutionary, most urgent changes we've ever seen in any industrial sector in human

history. So it's a real clash of cultures. LeGrand came in and we had other investors that wanted to come in and we pushed one out. They were pretty mad at me because I wanted to keep the cap table relatively simple. We have to manage a board, right? And with too many competing factions, it just becomes hard and hard that. But also we wanted to focus our initiatives like we could make our five year plan just putting effort into John's controls or just putting effort into LeBron. We also have a relationship with Virta and others. So I'm very careful about how to manage our bandwidth as an early stage company and make sure what we do we execute well on. But legrands, how many? No, a lot of people don't know. It's about a \$7 billion revenue company, about a 40 billion dollar market cap. They come from the power conditioning side in the white space. So within the racks. Power distribution units, Busways. They're a very acquisitive company. They bought Starline, they just bought a company that does a little bit of great race based medium and low power conditioning out of Colorado just about six weeks ago. They have not done minority investments before ever. So it was sort of interesting when we started this. They're like, we don't know how to do this. And it was clear they didn't know how to do it because their lawyers and their whole diligence process was based on acquisition. I mean they wanted our employees Social Security numbers. And I'm like, that has nothing to do with you. Right? I had to keep in mind, kind of stay in your lane. You're making a minority in, but you're buying like 5% of our company. You don't get all that stuff right. But they are on the other hand incredibly enthusiastic and aggressive, which we wanted. Now what we're working on with LeGrand is they were going to market with them. We have 30 accounts that we are working jointly on. These are typically accounts where they are wall to wall the grind. So their salespeople know the decision makers and they're bringing us in on the thermal side. And now we're obviously in there with them. We're kind of selling shoulder to shoulder right now. Eventually they'll ramp up to be enough knowledge transfer.

Speaker F: They'll sell is their revenue Europe, US or global?

Speaker A: It's about 40% Europe and some in Asia. And most of the rest it's about 30% North America, but they're known here. I mean, I'll give you an example. They brought it, the high frequency traders. You know, this is a really good example actually. Hudson river brought us in. They're interested in our two phase cooling. We found all the grandma's in there. We start selling together. The sales months are just accelerated because you have to understand that Hudson river is looking at us saying we love your technology. We don't really know you guys. We don't. And we turn to LeBron. They're wall to wall LeBron. So like, well if LeBron's backing you, that gives you some, some credibility. So we're doing a lot of matchmaking. You can imagine a lot of concerns about scaling, a lot of concerns about support, support service. We're addressing a lot of those issues with talking points, but also with these partnerships. The other thing that the grand relationship does for us is

there's going to be a convergence to power and cooling in the white space. So if you're bringing in that much power to racks, that is creating heat in itself, not just the GPUs and CPUs that's creating heat. So we're talking about developing liquid cooled busways for Legrand. We're also, we've agreed, a joint development effort which will be a kind of a prefab compute pod with all the power and cooling integrated into it. And I think this is a really good approach. We can talk as much as you want or as little as you want about the AI sector. It's hard to talk at all about it without sounding a little provocative because everyone has a perspective on it. I have some pretty strong perspectives on it, as you can imagine. But the one thing that seems clear is that there's all these. It's sort of an arms race on these large Data center builds, 1 gigawatt, 2 gigawatt, I mean every week we're approached by 7 gigawatt. Well, no one has this power. Power doesn't exist yet on Earth. The lead time for natural gas generators is about 24 to 36 months right now. So the trend now and the use case for AI seems to be moving toward slms. So small language models that are on prem. The term sovereign is being used. Sovereign a year ago meant it's a national security AI sort of implementation. Sovereign now sort of means air gapped within Coca Cola or Citibank or JP Morgan. We absolutely believe that's going to be the trend because the valuable data is the enterprise, the private data, the Internet's Been scraped already. And the other thing, and that comes to us because you're talking about integrated pods of compute with cooling issues. So if it's on prem, you're going to want an advanced enclosure that removes all the heat from that. You don't want to have water loops, you don't want to have any of that. So that kind of plays to our strength. The other thing that kind of plays to our strength and McKinsey called this a couple years ago, they said about, from 2027 on, the growth in AI is going to be inference pods. So not large language models but smaller pods that are closer to the people using the application. Right. So self driving cars or whatever it is because latency becomes the issue there for LLMs, power is the issue. So you want them close to power sources, you want them scaled out for inference, you want them close to the consumer, scattered throughout Austin or Chicago. Now where that lands we don't know. Like how, how important is latency? There's been improvements in, in that as well. But we, we certainly believe that there's another force at play here which is community protests. And it's much easier to implement a 10 megawatt or 5 megawatt data center than it is 1 gigawatt data center and fast drop or something. You know, where you don't have water, where you don't have the energy, if you get the energy, you're going to face residents protest because their energy bills are going to go up. All those things you hear about. There's about \$200 billion of data centers that have been stopped or stalled because of community protests or moratoriums right now. So every day in the news you see Ohio, Virginia, you know, having New York State having moratoriums on data center fields. Ireland, Singapore. It's a global phenomenon. So far so good. Is this going into too much depth or is this, it's a context that kind of, that we feel like we're constantly navigating.

Speaker B: So if I understand you, the concept is to deliver sort of a plug and play type solution for the customer. And, and is that something like the Dell provides or like super smci? Like is that the same type of

Speaker A: product that they provide? Yeah, so I'll qualify the answer and that we have to play in a lot of places. So the, what we call hyperscalers, the tier one, Google, Apple, Meta. There's a weird dynamic in this market market that's founded on another weird dynamic that Nvidia. There's a huge concentration of power in Nvidia. If you Think about if you map the profit pool, Nvidia would be 80% of that. Right? That's never healthy. IT infrastructure. The markets always found that impulsive. Typically it's corrected over time. I think this will be corrected over time too, but it would take some time. So Nvidia then says, who are my big customers? Because I need to allocate GPUs which gets into an adoption issue that we can talk about. So I'm going to allocate all of my GPU, 90% of my GPUs to these top hyperscalers, right? Because they're taking up all my back, all my, all my manufacturing capacity. So the little neoclouds or the little COLO players don't get any allocation. So it's a pretty dangerous concentration of power because they're actually shaping the market through that. Right. So but we have to, so we have to play with hyperscalers. So we are playing with the hyperscalers now. We have contracts with two, we have an arrangement within the third and basically we're playing with them mostly on this disintermediation play. So they're developing their own silicon now and so they want us to thermally, they want us to cool their accelerator chip, their TP like sort of chip. So we're doing that with a couple of the hyperscalers. But we can show you some of that work back there but that they're very heat dense, high heat flux processors. So it's kind of interesting for us. So we're going to continue, continue to play with that. But we think our sweet spot is in the sort of appliance based model. And Dell will sell them, but we will engineer them for them. We and our partners will engineer them for dell or smc etc. So I think you're going to see a lot of that in the VERTIV context or vertiv where oem, one of our products will show it back there. We just, we're just, we just launched it a couple months ago. We think the most success Verta, if you hear about Vertiv's growth and their backlog, all that's direct sales to the hyperscalers. It's all hyperscaler focused. So we think there's a play with vertiv to go through the channel. Have a Virta branded appliance connect with a server OEM and say put your servers in this, the channel can sell this as a block. Your channels like to, we used to call the channel box bloggers in the UK because they like to sell boxes, kind of, you know, simple blocks. And we think we can orchestrate that for the channel. Does that help? Yeah, kind of a long answer.

Speaker B: Maybe I'm jumping ahead, but are you concerned at all that a company like Verta that has just tons of tons of money just come in and replicate the technology, steer it from under you?

Speaker A: Yeah, absolutely. I mean I just, you know, part of my job is to be paranoid.

Speaker D: Right.

Speaker A: And there, you know, we have a good patent portfolio but you know, two phase physics, thermal dynamics is a lot of prior art. I can't imagine we go to court and defend a patent, but I think moving fast. So when I last started was a SaaS platform and all you, we had no patents, it was just moving fast. Right. I want to bring that to Excelsis too. I think we move fast, we enhance our performance, we're clever in the way we choose what platforms we actually spend money developing and we're more agile. Vertiv right now is growing well, their stock hits a new all time high every week, I think every day. Every day. I don't own Vertiv and I kick myself for that. And I keep telling myself it's too late. They divide Vertiv and it never seems to be too late to buy Vertiv. But I do think that they're not loved by their clients. Their value prop is we do everything. So back to my point about sort of comprehensive architecture, Schneider and Virta, their real proposition is we can do power, we can do cooling, we can do everything. So you have one throat to choke. And for data center operators in a hurry, that's sometimes helpful. The fact is they don't do a very good job of actually delivering. So with all the big company, I think we can be more agile, we can be more reactive to the market. You know, just I'll tell you this product that I'll show you in the back that Virta's gonna put their label on and sell, it was basically developed months ago, but then it has to go through their new product introduction process which is like 18 months. You can't move fast and do all that. So it's, that's a really traditional big company, small company. If we can't move faster than the big companies, then we really don't have the right to exist. Exactly.

Speaker G: It doesn't make any economic sense, right to develop it. Right. It's a great question that you ask, but it just doesn't make any economic sense that we're working with you just how fast the technology is moving.

Speaker A: It's moving really quickly, but every. So the foundation of all this conversation is that everyone knows two phase will be the dominant cooling architecture in X number of years. No one disputes that anymore. No one. Nvidia admits to it. Everyone admits to it. So Vertiv is working on two phase technology in the lab. Cool it. They just sold to Ecolabs is working and has been. We have dinner with them once a year and KKR is working on two phase and has been. Two phase is hard and that's the natural sort of technology moat around us. Not so much our patents, but just it's hard. That's why we have six PhDs on our team. It's complicated to simulate, to orchestrate and then to mechanically engineer it safely. Right. Because every CD we have is potentially kind of a bomb if people can it. Right, Right. So it's a pressure vessel. So we, we, it's difficult and we think we have that going for us. So Virta tells us they're about two years behind us. But will they try to steal our

technology? Will they try? Yeah, absolutely. Right now there's sort of this innovator's dilemma. They have 24 months of backlog on single phase water. They can't scale their factories fast enough. Nor can Schneider with Motive Air. So they're like. Well, they're sort of part of this party saying we don't need 2Face. We don't need 2Face because their whole team is doing this. So with Virta, very consciously we've said let's make this a channel play. We'll get traction in the channel and then their direct hyperscaler sales team will say wait a minute, this seems to be moving. Is there some applicability? And of course we're not going to give are hyperscaler to Bird of So fair enough. And tell me if I'm just.

Speaker G: Are there any headwinds at all right now for the two phase that you

Speaker A: see, there's a lot of headwinds. Yeah, I mean, I mean if you live in our world, you feel like you're beating your head against a wall and I'll, I'll give you an example. So I'm doing a. I'm doing a. So let me, let me lay out a few things. One, everyone knows it's going to two phase. Everyone knows the advantages of two phase. There's not a sophisticated thermal engineering shop and that includes Microsoft Meta, all those. They don't know this is coming. Vertiv knows it's coming. They're trying to lengthen the lifespan of single phase water because they've invested a lot in ramping up factories to build single phase CDUs and develop the skill set to deploy, which is really difficult. Single phase water sounds simple, it's really hard to deploy and the market's just catching up to that fact. But we have Nvidia who's not going to publicly talk about two phase cooling for another 18 months. They told us that. So we were in the Titan lab at Nvidia, which we've now been, we've been invited into and we're working on what that architecture is exactly, because they know it's coming. But the head of thermal for Nvidia was giving us a little tour and three weeks ago at GTC at their GTC event, they invited us to come over to the building to do this. And it was about seven or eight of us. Two or three of us were from Excelsius and then another water cooling company and then a couple of national labs. And we go through these mostly vertive equipment, water based and the head of thermal is saying, well, we have to switch materials on the piping, on the loops because the flow rate on water is already sort of untenable. Because in water you have two levers. You cool the water or you increase the flow rate or you do both. Those are your two levers, both of which are bad. Flow rate has implications. I'll talk about temperature is an energy suck. If you need colder water in the data center, it is hugely costly. So we can talk about some numbers in here. But he was saying, well, the water flow rate now is so fast that the single phase water loops are eating themselves. So they're actually leaching nickel out of stainless steel and creating particulate contaminants which are clogging microfitting structures and bringing servers down or making them therm trip constantly. And then you get into this maintenance thing where because water has gone bigger, bigger, bigger CDU's, right? Like 2 meg, 3 meg, 5 meg CDUs, they're supporting entire rows of compute. If three servers are clogged, you have to

bring the whole road down, right? It's a real bad business because you don't know exactly. There's no indication exactly what's happening. So you have to bring the row down and flush it. Verta just bought a billion dollar flushing company. So you see this happening with water that there's a whole cottage industry that's formed around supporting it because it's not an appropriate architecture for cooling water. Testing filters, a couple thousand dollars each they were designed to be replaced about once a year. They're being replaced about once every month. It leaks. You know, obviously water leaks. You've destroyed a few million dollars of computer equipment. If we leak, we don't. But all these things are real now. They're, they're, they're becoming real. So you would think easy, but you still have a lot of people that say Nvidia is saying we don't need it. And they don't support, they don't support 2Face. And we're going to go with Nvidia's reference design. And you also see, remember I talked about this conservative culture versus this revolution of technology. They're saying, I'm scared, I don't know what to do. But I know Fred and Sally, my friends in those data centers are deploying water. So it really does get back to no one got fired for buying IBM, which is a challenge I faced when I was at Dell. Like, we can build that supercomputing for you for 10% the price of the Gray or the IBM MVS mains. We can do that. And it's like, well, we don't believe you. We don't believe you. So we, it's, it to me it feels very, like a very classic tech adoption sort of wrestling match.

Speaker B: And why, why, why is Nvidia not supporting it?

Speaker A: So I, I, I asked him, I said, why are you going through these acrobatics to lengthen the, the lifespan? He said, josh, you know, because he was on a panel discussion with me when we founded the company and he proposed a two phased architecture and he was almost fired for it. And he said, we have to lengthen it. We have friends like Verde and Schneider that have invested a huge amount. We need to make it last another two to three years. Even though again, it doesn't make any sense in terms of math or physics. We have to do it. And he said, if I publicly speak, if Jensen came out in a keynote and said, we need two phase, we're going to need two phase, and two phase is better now, your stock would drop by 30%. There's no global supply chain yet for two phase. So we face these things every day. And the real crux of the issue now, and boy, we're still on slide one,

Speaker G: is AMD saying the same thing that it began.

Speaker A: AMD is hungry, so they want to differentiate themselves. So by the first week of June, we will have a consigned Helios rack, which you'll probably hear about. I think Meta just bought \$100 billion worth of these servers and these double wide racks with AMD GPUs. They are, they want to start with both two phase and single phase reference to designs. We will be the two faced reference design. And they don't have the room in Austin or the power, so they will bring their clients to our facility to look at their reference. Which is

I think a real strategic coup for us. I really like that. But, but it is still frustrating. The other frustrating part, that is probably what we face every day. And it comes from the Nvidia point of view. We go to Dell and we say support 2 face. And they're like, Nvidia doesn't want us to. We're like, we can give you a couple hundred megawatts worth of deals. Yeah, we're not interested. And we're not interested for one, because of the Nvidia factor. Two, because there is this belief that the plumbing, the plates and the plumbing within the server are completely unique to it. Single phase or two phase. So Dell, you know, the proliferation of SKUs is anathema to Dell. They just, they don't want more SKUs. Right. So we are now talking about universality. So I put a blog out three, four weeks ago. It got a lot of attention. I wanted it to be a grenade out there because as a small company you have to be provocative. But basically what we're saying is, is that you're making a big deal out of this. And I ran this by Nvidia and they're like, what do you mean? And I said, what if we can come up with a server architecture? The plates, the plumbing, the brazing, the manifold structure within the server, all that you can plug into single phase water or you can plug into two phase. What if we can do that for you? And they're like, what? I mean, no one's thought about it. So we're doing some testing now to show because our first test we wrote a paper with intel saying you take a single phase water cold plate, we actually perform better on that, even though it was designed specifically for single phase water. Now we're saying, what if we test entire server assemblies and we can prove that we're as good or better. There's some elastomer compatibility issues, but those are easy. Right? And then Dell, you have one SKU and then you have client choice, which you always talk about. Isn't that wonderful? And we've gotten a lot of good conversations on that. So we're partnering with Delta. Delta is a skilled Taiwanese manufacturer, mostly plates, heat pipes, etcetera they're one of Nvidia's prime partners to design their in server thermals and they also are one of Dell's big suppliers of cold plates. So we're working with them to do this testing, share data and then we'll probably come up with a joint proposition with them. And again, we need a partner because Dell's going to say, well that's interesting. Academically you're an early stage company. If we go with Delta, then suddenly we get a lot of traction. But it is still internally adoption is, we talk about it every morning. Adoption, that's what we talk about. It's frustrating because the other thing I wanted to say was in DCW next week I'm doing a presentation as part of a panel discussion with Vertiv AMD Supermicro and I think Cadence that does a lot of digital twinning. And what I did, I thought how can I, because this is our fourth year doing this, how can I make it kind of more real, real for people? So I went to chat GPT and I said tell me. And I refined the prom. You were all used to doing this, right? At night over a scotch rabbit hole in other stuff.

Speaker B: My new best friend.

Speaker A: Exactly. So I said in terms of, you know, academic papers, data center architects, everyone tell me, do a compare between the pros and cons of single phase water and two phase cooling in the data center. And it listed about 25 attributes. This was over a few iterations of refining the prompt, but 25 attributes, two phase cooling is green in all of them, single phase waters, red in all of them. So then I asked Chad, I said, so I'm building an AI factory or an inference pot. What do you recommend I use for cooling single phase water? And that's exactly where we are in the market because obviously Chad is a repository of data, but it's also a reflection on all the conversations going on. Everyone knows two phases better, but I'm still gonna buy IBM, right? So I'm gonna present all that to this crowd and see what happens. It's going to be another provocative conversation. It'll be good.

Speaker B: So you said Nvidia is about 18 months away from giving the check mark?

Speaker A: Yeah. So let me walk you through a couple of these. So this is the, this is from ASHRAE Semianalysis. Semi analysis is one of the most credible sort of deep technical analytic firms in the AI age. So basically there's this sort of consensus at around 1500 watts per socket. So per GPU, think of you have to go to two phase, the same was true. Remember, 400 watts, you have to go to some form of liquid. These are very fungible. You can increase flow rate, you can decrease water, you can do acrobatics, you can do a lot of things. But basically thermodynamically is where you want to go. No one really argues for this. So if you think about the AMD Mi450, it's going to be about 1700 watts. And that's coming out in the next two, three months. If you think about the. And they're already. We're designing the thermals for the Mi 500 series with them. That's actually going to be what we display a simulation of those chips. Those are going to be about 2200 watts. So you see that the forcing function, if you get into academic arguments or not about is it 1400 watts? It doesn't matter because the curve is exponential on the watts delivered and the concentrations of heat, which is really what matters. If you talk to thermal engineers, they're like, I don't care where the heat is because you might have a huge surface area. What I care about is the watts per square centimeter. And the heat flux is going up high as well, beyond what water can. So it's just, it's going to happen. We're better now, required later. That's what we always tell everyone. But it doesn't mean these adoption things go away. So here are the differentiators after this slide, I'm going to get you into some market data. So we have sphere heat. What we mean by that is we can remove heat more efficiently with our phase change and our fluids. But also we don't really care if it's the Rubin, Rubin Ultra Feynman whatever chipset. We haven't really seen a limit on this. And that's why every industrial sector goes to two phase, you know, your car, your house, your, you know, nuclear thermal blankets for nuclear power plants are two phase cooled. There's really no limit to this, to the capability here. So it's kind of future proof. The one advantage,

one thing that was thrown at us was tco. Your total cost of ownership is going to be higher. That's what the water guys say. They have a bunch of PhDs. It's complicated, they admit it's complicated. So therefore it's much more expensive. And they're using custom componentry for their CDUs. So the supply chain will never catch up to our supply chain. Right. The fact is we use mostly off the shelf componentry. A lot of it's the Same componentry, this cdu, this single phase guys use. And so on a CAPEX basis, think of this about the same. On an OPEX basis, we're about 35% advantaged because we can use warmer water loops. You don't need. If you design from the ground up a two phase cool data center, you have all kinds of capex and OPEX advantages. You don't need all the chillers, you just don't have to buy them. You don't need a lot of that infrastructure. If you already have it and we plug this in, you don't get that advantage, but we certainly get the future proofing advantage, etc. And you can, even if you design it for that and you use two phase cooling in your IT sector, in your section, you can warm your water loop by many degrees and for every 1 degree C you warm your water loop or you don't have to cool your Waterloo, you save about 4% energy a year. Now if you have a 10 megawatt data center, it's kind of who cares. The sectors never cared about efficiency at all. Mostly during players who take kilowatt hour pricing and just put their margin on it. So it literally is almost a disadvantage to being effective to be efficient. Now if you think about 500 megawatt data center or 1 gig data center, you could power Cincinnati with the energy you saved by using this. I mean the number, it's almost beyond comprehension, the scale of these data centers. Fair enough. Non conductive fluid. So if we do have a leak, we probably will have some leaks because a lot of humans involved in this, it doesn't damage anything. I could put this notebook in a bathtub of this fluid and take it out and use it. It just, it's a non conductive fluid, it doesn't affect electronics.

Speaker B: And that's proprietary.

Speaker A: No, no, consciously we decided we don't. We're not chemists. I mean my CTO is actually a PhD in chemical engineering, but we don't use them for that. We use them for heat transfer, but no, we want global availability, we don't want to worry about certifications, etc. So interestingly, Honeywell Solstice SK is South Korea. We're developing a JDA with them to both use us in South Korean data centers, but also use us to help test their next generation fluid. Kimura, the CEO of Chemours, which is a spin off of Dupont, was down here last week with their team all day and at dinner. They are so interested in this and we need them because remember we the Regulations, you know, going through EPA or eu, that's years. I mean these things have long lifespans and we don't want to take that burden. But also right now our equipment is sort of enterprise. You roll in, plug it in, you know, a couple hundred pounds of refrigerant, it's not a big deal. Some of the deals we're working now are for blocks of 15m kind of pods. Now then you're talking about thousands

of gallons of refrigerant. So we don't use a vacuum to vacuum it. We don't use, we can't charge it. You need trucks, you need vacuum trucks, you need vacuum tankers to support that. Kimours, particularly Kimours wants to do that service for us and they want to go into a different data center with us and certify that it's safe. So if there's a leak, you got ventilation there, you have oxygen tanks there, you got all this stuff which they're, they've been living this for years. It's all new to us, right? So instead of us making reassuring noises, they come in right with us and they say, here's how we support this infrastructure. Awesome, cool. And then serviceability. We'll talk about some of the service that's an except not two phase, that's just a tenant, an engineering tenant that we use to design our equipment. We'll give you some examples of that. But the infrastructure compatibility one is important. So most people are building data centers with single phase water in mind. We can plug and play. So it's important that you all know if you build a water loop and a chiller infrastructure, we can plug, we can sit right next to a water based rack, it doesn't matter. So there was a concern, it was a narrative thrown out about a year ago at us from the water guys saying you're going to have to throw away all your capex that you invested in the shell build. You don't have to do any of that.

Speaker F: Why would you work together? Why would the data center, why would you use both water and two phase? Why wouldn't you replace water with two phase?

Speaker A: Well the important thing here is that we're on a journey and so what we like to go in and say, listen, you know what's new? You want to start playing with this technology right now. So if you have a bunch of rows of IT equipment, put us in one row. So that's why there are two reasons why we push that. One, we obviously want data centers to be built from the ground in two phase. We're just not there yet. So we have to sneak into the data center. Centers are being built.

Speaker F: Right. So where they're placing yours, they're turning the water off.

Speaker A: No. So the, the typical.

Speaker F: Okay, not right. Not the whole data center, but.

Speaker A: Well, we don't. We don't. We. We would roll ours in with our CDU so water would never have been present in that rack. So you're just dropping in. We're dropping it in. We're connecting to the prime. What's called a FW facility water loop. We're connecting it to that same Waterloo. Right. That's what we mean. So let's talk about the market a little bit here. So I just hit on a couple points here. So the announced. So if you wonder how we can build all these data centers that you see headlines on, the headlines are the in 2020 of the blue, about 14 gigawatts of data centers have been announced. Those may take two to four years to build. So we get kind of lost in the headlines here. The actual prediction is that we'll

probably commission about 5 gigawatts of data centers this year worldwide. And that continues to lag well behind the announcements that you see or the build starts that you see. We simply did that to kind of ground ourselves in reality. And then look at. And this was over scotch as well. Kind of a rabbit hole that I did last week. But the interesting thing is if you go from top down, it triangulates almost perfectly with what we've been assuming and the forecast we've been working with. So a couple points I'll make here in 2024. It's a billion dollars of liquid cooling. That's all water. It's a tiny market. Billion dollars. So you hear about water, Water, water, water. Well, it's not an incumbent. The narrative is that it's the incumbent. It's certainly the first mover and the dominant architecture. But the last survey we had is about 6% of deployments had water, liquid cooled with water. And the average size of those deployments was about 3.7 million. Meaning they're kind of proof of concept level designs. Most of that is hyperscalers. The other dynamic is where who's doing this? It's all the hyperscalers. Coca Cola doesn't have water in their data center.

Speaker E: They're just forced air.

Speaker A: Forced air still and to go with higher power racks, they just depopulate a rack. They're not doing AI in their data centers.

Speaker B: But Rubin requires liquid cooling, right?

Speaker A: Rubin requires liquid cooling. But the Rubin allocation is all going to the Hyperscalers, right? Or 85%, it's going to the hyperscalers. So then you see this growth of liquid cooling, which pretty much matches what we had assumed a year ago. Just looking at kind of tops down. And then you get into this. And so my point here for you investors is that despite the struggles that feels like a cage fight we're going through for adoption. This is a really optimistic kind of, this is a great market to be in. So we have the total liquid cooling market, which gets up to about 30 billion. That's just server assembly. So plates and CD is in the two phase market. By then we'll be about 9 billion. So very much in line with what we've always talked about. But the adoption frustration we're having is look at 25, the total two phase market is about 25 million globally. So it's just rats and mice. It's POCs out there. And then in 26, our plan is 90 million on \$150 million market. So still small. We take a large share of that, but then the market grows and our share declines. Based on our current pro forma. I don't want to give any of you the indication that we're just going to see market share willingly. It's just, I want to show you that despite the frustration, there's a huge market for Excelsior out there and there's enough adoption to feed our pro forma for the next two to three years before a tipping point occurs. Does that make sense, Josh?

Speaker E: So this projection is kind of base case for us?

Speaker A: Yeah. Yeah, I think so. What you'll see though, the 90 main. Is it 90? Is it 50? Is it 100? You know, you'll see, you, you'll see that kind of has some volatility, but you can see how conservative our pro form is based on this. Now, the fact is, you know, 27 to 28, there's no way. I think maybe I should put it this way, just based on, I mean, Diego and I were comparing notes. We probably had 10 companies saying they're going to buy us. And all I can do is shrug it and smile at them. I don't know what that means. Okay, you're going to buy us. We had one yesterday that just sent me a note saying we want to buy you. And then they're back by taking their back by taking. They have deep enough pockets, but we don't think we're ready to sell. We have a lot of value to create, but by 27, 28, there's probably going to be an offer we'll have to seriously consider would be my guess. The other thing to think about is the number of independent dance partners in liquid cooling. You know, Chill Dime is bought by Daikin Motor is bought by Schneider, was bought by Jabil, but Jekyll was bought by Flex Cool. It was bought by Eagle Labs. We're kind of, kind of alone here and I think we're the prettiest girl to dance. So someone might jump ahead and say we're just going to grab you. We'll see. I have no idea.

Speaker D: What's the implied valuation based on those comps for Celsius in your opinion?

Speaker A: Well, right now that's the thing. That's why I don't want to sell now. Right now it's based on our team and our technology. We don't have a lot of revenue history. What I want to get is some revenue history. And the comps right now are easily 10 to 12 times revenue. So if we made 90 million, made a one set 171, you can do that.

Speaker F: Is there any line of sight today on when future data centers will be built only with two phaser they could extend that Runway five years.

Speaker A: And this model looks even if you the insiders at Nvidia who won't say this publicly, they'll say by 2031 no one's going to be building non two phase data centers. Everyone's going to build two phase. So that 30 billion kind of TAM I have up there there becomes our SAM. Essentially right now I want to be conservative. We want to focus on building a good business and setting expectations. If something great happens, then it's great. If an exit happens, then an exit happens. But we're not constantly looking for an exit or any of that. We're completely focused on just executing our plan.

Speaker B: If Jensen comes out in let's say 18 months and says we endorse, we think the future is two phase. Why in 2029 is two phase only 20% of the market? It would seem like it would be

Speaker A: so much bigger because there's still a lot of traction. There's still going to be a lot of resistance from 100 vendors of single phase. I will tell you what will happen if Jensen came out whenever and said said what we say it's better now. It's required later sewn by the

following unstable bias. Yeah, that's what's or make us an offer that women struggle to refuse. Right. So as of today you're leaning toward a sell versus going public in Theory. I'm not leaning toward it. Well, I'm not. I mean, I'm too old and tired to run a public company. Sure, I've done marine sculpts before, so it won't be me, but. And that's only half joking. But what I'm laying out is what I think the probable scenario is. And I think the probable scenario is if someone's going to make an offer that it would be silly as investors not to take. I don't know what that is. Is that a billion? Is that 2 billion? Is it 3? I don't know. But there's going to be a point in time when I think someone's going to confuse us with an offer.

Speaker F: But you also said the lag time is at least two years to build a data center. So architecturally, if you haven't even started

Speaker A: and you know,

Speaker F: in the next generation

Speaker A: of Nvidia chips, right,

Speaker F: why wouldn't you just automatically start designing with two phase? Like why would you even consider. Because single.

Speaker A: Because Nvidia is saying we do not endorse two phase.

Speaker F: Yeah, but prior slide and you discussed previously, the laws of physics demand it. So who cares what Jensen says? If you're looking at the laws of

Speaker A: physics as governance, isn't that confusing. This is why we're, this is why we're banging our head against like it's so. I mean, first of all, let's step back a second. We have a differentiated technology which we know is going to be needed. We're the market leader in a really high growth sector. It's kind of a cool position to be in. The frustration is, you know, this is true. If you talk to Ali Hidari, the head of thermal at Nvidia, he's like, yeah, we should never have gone down this road. It's stupid, it's bad, it's destroying GPUs. Right. It's costing millions. By the way, I posted some on LinkedIn. A couple days ago, Bloomberg had an article on insurance premiums in the data center. I apologize, I had a typo on that too. I should always have someone proof my post. But Bloomberg said that, you know, aviation's always been the biggest sector in insurance for premiums. Data centers taken over and by 20, 2030 it's going to be just under \$25 billion premium market for data centers. And then the article goes on to point out the reason for that are thermal architectures which destroy GPUs. Right? It is for us it's a little Twilight Zone ish. You know, it's like we live in upside down. I mean, look at the math, look at the physics. We have a viable product. Can we scale? Yeah, we can, we can scale. We got CM Partners, we got John's, we can scale. It's just that everyone. What I tell my team often is just because a lot of smart people are in

group think doesn't mean they're right. And that's what we're facing here. A lot of smart people are saying single phase water is fine. They know deep down it's not right.

Speaker F: But if you go backwards in time, I think it was about a year to 18 months ago on one of the inventor calls. I'm not sure who it was who really laid out a lot of granularity around your pilots with hyperscalers is that you listen.

Speaker B: They're also the angle with the TPU, you're working with the TPUs and they're quick to adopt two phase. Then it sort of proves that it works and then it makes it easier because if the customers are the ultimate decision makers, Nvidia will do what their customers ultimately want, not what their vendors want.

Speaker A: Well, customers are buying. Customers don't really love Nvidia right now, so I don't think that's really true. Do you think?

Speaker B: No, I'm saying like if, if you could prove that it works well with TPUs, if you said you were working with hyperscalers on that, now the market is like, okay, this is a proven technology, it works, then it should be adopted quicker.

Speaker A: Yeah, it should be. I'm just, I'm just trying to be realistic and lay out there's a real strong force against us right now. As illogical or rational as it seems, when Nvidia admits to me that it's irrational, they admit this is a bad architecture. We should have gone to two phase when we started this stuff. You know, it was a misstep, but now we're stuck with it. When they admit that and when I say, are you gonna, when are you gonna talk about this? We're not gonna talk about it for at least another year. We're not Right.

Speaker F: But, but the hyperscaler doesn't have to care what Nvidia says. They just have to look at the laws of physics, which goes back to the pilot.

Speaker A: That's why the hyperscalers. But remember, in the hyperscaler world, Microsoft or Meta, you have multiple data center architecture teams. You typically have one who is deploying today and they're deploying as fast as they can. If I went to them now, they'd Say you're complicated my life. I'm going to deploy with the plan of record. They literally have a plan of record, an architecture of record. That's what they're deploying. We're engaged with the future architecture. Teams who know this is happening. They have to test us. They're very concerned about scaling with quality. And also there's just a lot of. There's a lot. Selling the hyperscales is not a fast thing. Right. A two year cell cycle for hyperscale is pretty fast. Right. So what you're expressing is absolutely the world we're living in. You know, the data shows we're better. We have equipment that's actually deployed that is working the way we say it's going to work. We all know we're gonna need this probably sooner rather than later. Right. We all know that flow rates are already untenable and they're already damaging themselves. So why aren't we moving faster now? I think we'll look back in a couple years

and say, wow, that moved fast. It's just that we're living in it right now. I think we'll move, we'll look back and say, wow, it was about.

Speaker E: So don't you think, Josh, like if

Speaker A: I go back to.

Speaker E: Of hyperchale adoption, everybody was in kind of that same mindset.

Speaker A: Yeah.

Speaker E: And then you had Morgan Stanley who was very aggressive. I kind of view them as kind of the trendsetter in next generation data center architectures had a success and that was broadly adopted. Is there potentially a breakthrough opportunity with somebody who wants to pierce through that membrane? Is AMD potentially that beachhead or.

Speaker A: I think AMD is potentially that. And we certainly will get. Will be promoted. That will be promoted and people will take notes from the Nvidia will notice. Right. I think from an end use point of view, I think our biggest credibility play is going to be the Fintech and high frequency traders. Everyone knows they're incredibly sophisticated. They have incredible IT teams, They're good technologists. When they adopt us, which I think will be this year, then we'll see. So we have basically kind of a roadmap of we always think about building momentum. I don't know if you thought about this consciously when you were doing brocade. You know, if we do this and then we do this and we do. Can we. You're building momentum that we're obsessed with thinking about it.

Speaker E: Well, this is like a back to

Speaker A: the future for you. Yeah, I can imagine, I can imagine. But it's, you know, it's pretty Exciting. And I think the frustration we and trust me, I mean we have talks amongst the teams like come on guys, it's one battle after another now. But this is. We're going to get there.

Speaker E: It's kind of interesting the strategy that you talked about about the prepackaged systems. Is it conceivable that you could have broad adoption at that level and the enterprises are actually deploying two phase but don't necessarily know they are in their kind of trend setting.

Speaker A: It's so funny that we got in this discussion with there's a bu. It was an acquisition from Johnson Control, Silent Air and they do modular data centers and we got in this discussion they're really good high quality people and they said, I said do you think if we develop a modular data center with our cooling architecture and we're more efficient and safe and and you can pack more compute in that because of that. Do you think they care

what cools it? Like they don't, they don't even have to know. So it goes back to Dell has to know, right? Super Micro has to know because they have to, they have to partner. They have to get with us on this right to. So we're doing the universality thing we're also doing. But the Dell will tell you their, their engineering group said Nvidia's not doing it. I'm not going to stick my neck out. And you do see a lot of the market now that they're sort of empty suits. Dell's a weird one because you see a lot of executives who are up a year from retirement and they just want to code. I mean it probably reminds you of other companies at some point in time too. They that going on. But Super Micro they keep getting their own nonsense. So we are looking at some white box. You know the majority of servers are not sold by Dell or HP of these brands anymore. They're sold by no names. White box, Quanta, we win. Wistron, Foxconn, Hypertech. So we're approaching that now and saying listen, you can be part of this pod. The customer doesn't care that there's Dell's badge. It's going to be cheaper for them. They'll charge you and they're going to be more responsive to your needs. Why don't you do that? So one of the programs that we're actually presenting in our board meeting next week is that. Does that make sense?

Speaker E: Oh, total sense. So another question for you. How should we think about the commercial terms with JCI with Run is it reference selling is it co selling is it oem?

Speaker A: So let me take them separately. So legrand, we are co selling now and we started like the day after the investment was closed. They really want to get into this market. We are, we have a work stream on co development projects. So this advanced enclosure, they have a factory that already does prefab stuff. So it's a great partner for us. And we've started a workstream now in private labeling. They want to put their brand on it. So they're moving really it's over three months doing all that. JCI is a weird one in that they don't want to move until we negotiate a JCI and they sent us to JDA and it was completely unacceptable. It was like any co developed IP is their ip. You know, it's one of these. So I wrote back and I said listen, in the side letter of the investment we're going to move to a commercial agreement in good faith. And so I said this isn't good faith, this is nonsense. So they come back, we went back, we're going back and forth. We will in the end probably offer warrants to JDCI and LeBron. If we do one, we have to do the other for the operating agreement. The advantage of that is this that we could say if you generate with us through co selling, reselling, co development X millions of dollars in incremental revenue will give you so much more. More of our. You'll be will be allowed to buy so much more of our company at a discount. That's a great incentive for us because all the investors win. So you know, if they can generate \$100 million of income of revenue for us. Think of the market cap, right? It's multiple, pick a multiple. It's at least 500 million in market cap. So the warrants are actually pretty inexpensive for us and it ties them. So once we get this body of an agreement with JCI then we'll put the warrant appendix on that and we'll give it to LeBron

as well. And I think that would really speed things up. They were really concerned that you know, you can imagine negotiating the minority investment. I'm saying well you're getting part of the company, you get upside. But you know, if they really lean in, they're going to get, we're going to disproportionate upside. So I'm, I'm all in on warrants as long as the milestones are quant. They're real revenue, not qualitative and they're big enough to have a material impact. Does that answer your question, by the way? But under jci, just to finish that off, we're going to focus on the comprehensive architecture and we're going to focus on so actually build a modular data center and they're going to have to sell a lot of them to meet this target. Were either two of those your relationships or were they Goldman's relationships? Now you're just trying to get my blood boiling. We thought, I mean just because we're all friends, I'll speak. Amazing. Goldman courted us to do this feeder. Unlike what we're not going to move any of the gold. But in the end I felt it may have some halo effect. They may bring in someone that we don't really have access to. Maybe even Nvidia, you know, call them big game. I was talking to Johnson Controls, all the investors before and I closed the investors without their help. I would say they were negative value and that they had a team of ten 19 year olds who I was trying to teach and Diego has wasted hours on this trying to teach them. They would come up with a deck and I'm like, use my deck. And they would say, well we want to frame it this way. I'm like when I tell you messaging, I'm dealing with clients all day long. When I say a message and use words, I want those words used. And maybe I sound like a jerk now, but you had, you know, 22 year old, you know, three in the morning trying to add value. And so we kept doing that. In the end we paid them a fee. They did nothing but slow us down. They didn't bring in one new lead, not one. So I would never engage with Goldman again even on a big transaction. I tell you right now, I'm not.

Speaker F: On that note, do you see yourself doing a series C before potentially entertaining and M and a transaction?

Speaker A: I don't know. I don't. We plan this in a way that based on a number of scenarios, we have enough cash so if no one buys anything for the next 18 months, we're okay. If people buy. Actually the reverse is not so true. It's easily solved. If we needed a lot of working capital because we're growing so fast, I can find money, but we. The plan was not to do a C round. I would never say never because I don't know what the future holds, but that's not the plan right now. I think direct to some other transaction is probably better for us.

Speaker D: Josh, percentage wise, how full is your team that you have? And I know you said last time we met that you've got a good base of talent to pull from the local market. Are you still head hunting and looking

Speaker A: for more or are you still out? We're still head hunting. We have mostly engineers so we're bringing on a couple more mechanical engineers. Another PhD as accepted that will join in the summer. The only job class that we're struggling with are double Es. They're just really hard to find. So we're trying to look for another double E. But basically the talent pool here has been fine. Double Ease is the only one that we really struggle with. And so we're constant. We just decide we're constantly going to post for Double Ease and we'll find good people along the way. Our firmware software team is already using, you know AI so they can scale about 70 or 80. Code that we write is. Is written by Claude right now we did announce in our all staff every month. Every month we announce new employees and promotions and awards and you know, the typical stuff. The last month was the first time we announced two AI employees. Two agents with names.

Speaker G: Oh boy.

Speaker A: And Harper Snow and Henry Frost. And Henry were. Works his butt off for Diego. I mean we now do our weekly sales report as Henry does abs. We do. We get a weekly. I mean a daily news update. He distills the data center news and sends it to Diego every morning. Don't have that. Comes in our inbox from Henry. So we will supply chain marketing basically stopping hiring. Select operators would need obviously but procurement planning and we're just. We're really headlong into using AI to scale off. That's going forward as well.

Speaker D: And facilities. I know you've added more.

Speaker A: Are you still.

Speaker D: Is it short term? Is it flexible? Are you needing more than that?

Speaker A: Like how does that fill out? I hope we need more than that. So there's this and then manufacturing and supply chain. About five miles from here. It's about 30,000 square feet. You know we walk in. We walked in this and I was intimidated. Right. And now it's packed. We walk in that and it will fill really quickly. So there's actually space next door that we'll probably negotiate an option on. But that space will allow us to generate enough units to to meet our revenue goals for the next 18 months or so. Thank you.

Speaker F: I had a question from.

Speaker A: This is again.

Speaker F: I think it's at least a year ago.

Speaker A: By the way, terrible hills. Do you want to eat coffee, Scott?

Speaker F: Sounds good.

Speaker A: We'll stop in a few minutes and grab some coffee.

Speaker E: That's a great idea.

Speaker F: So on one of the adventure causes somebody said quote the pipeline includes potential nine figure orders from hyperscalers though it's diversified across hundreds of leads rather than concentrated customers including Hyperscore have been testing the tech for the past two three quarters building confidence for larger commitments. This lines with earlier Q1 25 so it's. It's a while ago one venture holiday cell cell deep discussions with hyperscalers. Where are you on those tests that

Speaker A: whomever was talking it depends on the hyperscalers. So I mean I indenture says what an entry needs to say in their quarterly calls. I don't I just to be very transparent I want to be a private company. That's why early stage companies are private. I don't want to do a quarterly earnings calls. They try to extract a couple data points to excite people. If it were up to me they tried to get me on those calls. I said I'm never going to do a quarterly call with them. I tried in one as a. As a favor. I never want to do that. It's just nothing but downside for me to publish any of that. Nothing but downside. But that's topic for another day. I will say that one of the hyperscalers we continue to test plates and refine and do all that. The other one though is moving the CDU conversations. So let's do now a proper POC with your CEUs. It's probably going to start with our Mr. 250. They're not going to use that in production. It's really just the next stage of validation. They're going to use our 2 megawatt CE which I'll show you. You. How are we doing on time? Oh not doing very well. Sorry, I've been talking too much. That's fine. I think we can speed up the throw. Okay so let me, let me show you just so in case. So this is our Mr.250. This are our journeyman product. It's certified, it's shipping. It's 250 kilowatts of thermal capacity. It was designed to have a rack on either side with sort of a the density of an NDL72 reference side from Nvidia. It's about 125 kilowatts on each side. It can support three or four racks of lower density. The fact is we're now going to recertify this for an Mr.350. We actually over scoped the design. It can absolutely support 350kW. We want to do that by the way we want to continue to do that we don't want to get out in the field and it performs at 80% of what we promise. We never want to be in that position. A lot of our competitors on water antifa side are make claims that then they get out and can't come close to sporting. We don't want to do that at all. That's just bad self inflicted reputational damage.

Speaker E: Josh, could you talk about the competition for a moment? You've got the potential for avertive to try to get into the space. Zudicore comes up consistently as the primary competitor. How should we think about them?

Speaker A: So Zudicore so far I'm still not worried about them but it does look like Mitsubishi Heavy Industries and invest in them is giving them some assistance on design and engineering. It would be better for us if they were better. Honestly it would be better for us if they, they're better. The founder of Zunicore wanted to meet with me about three weeks ago at gcc. So I had a beer with them. He wanted to collaborate and I said and in my mind because I'm a smart ass, in my mind I'm like you being capitulated. But I wish they were better. If they had a really good product line and we could go to a large client and say where are your number one suppliers? They're your secondary supplier that'd be fantastic. It really speed up the market. They're just not there yet. But we do hear they're getting better right? So far like Hudson river had been in bed with them, tested their technology, really want to go to two phase and they failed all their tests. That's why they came to us. Right. So there's a lot of that happening. I would almost rather Hudson river had a good experience with them and then they said we need a second supplier and then we'd get in and show them that we're better. I would almost rather because every even at Dell talking to a federal district manager who sells in the government at Dell he had brought Zudicore into like two or three dales and they didn't lose credibility for Zudicore, they lost credibility for themselves. So and then the competition I'm really worried about, I'm worried about Virta screwing us. I mean every tree has a tiger behind it it seems like right. You probably. I'm worried about Boyd. You know Eaton bought Boyd. It's a big chunk, \$9 billion. It's a lot of amortization. But Boyd does have some two phase expertise so far. It's all focused on aerospace and other sectors, not data centered. But there's no doubt they bought them because they have that expertise. They probably have 10 PhDs that are two phase heat transfer experts. They have a lot of resource. So those are the competitors. I really worry about that. We'll carve out a niche, we'll be fine. Again, our market share plans are pretty conservative but I worry about the competitors that have scale and already have deep relationships within the name. That's right. We're on any given day. This is our. We'll show you this. This is our IR150. This is the product we actually designed for Vertiv. We can sell it under our brand but there's. And a white label as well. It's a completely integrated. We have an appliance type rack for an SLM or an inference pod. All the shelves there are for servers. The cooling actually is the 200 millimeters on the right hand side. It's the first of its type, completely integrated rack. It's a standard size rack. Our next version will be wider and deeper because servers keep getting wider and deeper. So. And we'll release that probably beginning of Q4 and Vertig won't have access to that unless I want to grant them access. So I want to gain a little bit more negotiating leverage back in my core vertical.

Speaker E: So you don't lose any rack space,

Speaker A: you lose no rack space. So it's 42 use of rack. You can completely populate it. All the cooling has been sized and

Speaker B: is the installation time just much shorter?

Speaker A: It's super cool because you. You basically imagine this going to a. You don't need like a facilities team involved. Right. A lot of the complexity here is it used to be IT people that dealt with servers and facilities, people that dealt with, you know, air handlers and power distribution in the mechanical gallery. Now it's all blurred and so you get into these kind of convoluted conversations. Very different and different work worldview.

Speaker E: Yeah.

Speaker A: Right. But here, yeah. You can imagine this going to a wwt, having the IT integrated into it, rolled out into the operator's data center and with almost no deployment complexity. Pretty cool. And then let's see, this is. We need to do commissioning with two phase two. So this is our load bank. We've got a thermal simulation rack. Essentially it allows a data center to do. To test us. So if we say you can have water temperature facility, water loop of this of you know, 38C and still keep GPUs under their throttle point I want them to be able to test that. I want to show that we want to be very data driven and this gives you a route to precision commissioning. We do want to be in the commissioning market and we have a meeting next week with a company, Lagrange, that does. It's the oldest data center commissioning company in the world and the biggest, I think. So they're very interested in maybe OEM or reselling this product. So that's what I wanted to show you. I don't. I have a rendering of our next product line which is. So we had to address a couple things. One is can you scale manufacturing? Can you scale thermally? So like Microsoft would say, we love this, we want to start playing with it. But we're never going to use a 250 kilowatt CDU. We need a multi megawatt CPU. So we wanted to start immediately developing that. And that's going to be a. We call it Mr. 2K, but it's a 2 megawatt CDU. It will be bigger than that but not as materially larger and it will support a row of compute or more likely the way that's evolving, a couple racks in a pod. But we are developing a pretty. It's called the Overwatch system. It's a pretty complex system valving actuators. So if you have a leak that reports to a sensor, reports to the system, it shuts a valve off. So you can contain all of this. Because we now are in kind of industrial level stuff and we need to treat it that way. You know, if there's a leak, it's serious and people could die. And so we need to put that sort of radar into our engineering.

Speaker E: Is there redundancy built into. Yeah, every single point of failure?

Speaker A: Yeah. So there's. That's taken a lot of engineering because now you have a lot of fluid and so there's actually redundancy between two seat thinking as bookended. Or they could be proximate but support a row. If one fails, the fluid management system adjusts and the other one rests. Yeah. So there's a lot of. Conceptually it sounds easy, but in terms

of engineering, it's complicated. We actually are partnering with Omron old company. I don't know if you guys remember Omron, he's from KTM machines. But they do a lot of this safety regulation for big factories and meat packing and stuff like that. So they actually could almost take something off the shelf, customize it for us. And that would be our control box for the overloads system so we're not, we're not actually designing that in house. It's our first piece of movement. We're not designing entirely in house just because we want to move faster.

Speaker B: Can you talk a little bit about Dark nx, like where the project stands? Do they have funding in place? A tenant like and any color you could shed on that because it's very hard to find information on it.

Speaker A: Yeah. So they have, they're negotiating now with I think how many off takers? Eight off takers. So and these clients, they have funding? They absolutely have funding. We have a contract with them for 55 million. So we have booked to build 55 million with Dark NX. The I think what you're going to see is that that was all supposed to be deployed this year. I think it's going to be deployed throughout Q4 and in the next Q4, just like every other project in the sector. It's been delayed by. Not because of us, but other issues. But yeah, that's. It's very real and it's going to happen. It's our stress is when is it going to happen? Right. So. And then we have another big.

Speaker B: Do they have the permitting and everything in place?

Speaker A: They haven't permitting it. So they're multiple sites. They end up the first site already, which would be the first tranche and they're about to have two more sites stood up. So it's not years away.

Speaker B: Are they going to break ground with the construction?

Speaker A: Oh, they already have.

Speaker B: They already started building.

Speaker A: They're in buildings.

Speaker F: Okay.

Speaker A: So they partnered with a big real estate guy in Toronto and he's using his facilities and he's got enough power to start now obviously they're going to put a best system in. They're doing a lot, but we can start without any of that happening. So I think that will be. I don't know when exactly it's going to start. I wish we could nail that down, but it's going to be a fairly nice ramp for us. And then there's another big kind of new. It's not really. These aren't really Neo clouds, they're really kind of Neo Colos. But another company that

they designed this sort of. They call them nests and pods and this architecture. And they have how many off takers now? Looking at them? Around seven. Around seven. So all these things are going to happen and they're absolutely in bed with us. It's going to end. Their off takers, all these off takers are very comfortable with our technology. That's not a problem either. Now imagine, though, talk about connecting the dots a lot. Once these things start moving, then we go to AMD and we say, you want your servers in here, right? So you put pressure on Dell to put our plumbing in because we're going to give you 100 megawatts, right? So even for AMD, 100 megawatts is a lot for AMD right now. Right? It's not so much for Nvidia, it's a lot for AMD. So as some of these things get driven traction, then you're going to see a multiplier effect because we'll be able to bring some of these other parties in. And that's our approach. If there are other approaches, we're all leaders, but that's. You can tell how much we think about exploiting a little crack in the market and trying to weave an opportunity out of it. I just don't know another way to do this.

Speaker D: Josh, when we were here last time,

Speaker A: I mean, you obviously have your product

Speaker D: projections in, looking back, say from past 18 months. Where are you on that track?

Speaker A: Are you above, below?

Speaker D: And what positive or negative surprises have happened that you didn't foresee that could work?

Speaker A: Either way, we missed last year. It just. The adoption didn't happen. And remember, I blame myself here a little bit because we in Venture Villa pro forma, before I was even hired, and I was the first employee and I sort of shrugged and said, seems reasonable. And so we just kept it intact. And that's still the one you see. Now the fact is we've caught up with that. And so the 90 million 171, I think. I don't know if there's any timing issues or what, but we're on track for that. But last year we had a million in revenue. I don't know. I don't even remember what the plan was, but we did not adjust that plan. And frankly, I don't, I don't really, I don't want to be glib about any of this, but I don't. You know, we're, we're focused on execution. We're focused on a book to build, building that, that book of orders. The timing is going to happen when the timing happens. So sort of, it's not something I've spent a lot of time thinking about. I mean, adventure has to do their reporting, and so they, they're obsessed with it. I'm a little bit more. I'm a little. Because I, I'm completely confident we're going to pull something off here that's Going to be incredible. So how much cycle time should I spend on thinking about POS this week or different. Different approaches. Which is. I guess I could take to that, but that's kind of my

view then we're still. I think we're executing well. I think we have a great team. I think we have great technology. I think we're in the right space. I think we're thinking through some of these challenges in the right way. Probably not thinking of all the best ideas, but we're thinking of some of them. So it kind of. That's about all we can do.

Speaker D: Sounds like the competitive concerns with the Eaton entity has cropped up since we last spoke.

Speaker A: That that's something.

Speaker D: Keep your eye on.

Speaker A: It is, yeah. Because Boyd, you know, Eaton bought this because they're a big power company. They have no thermals. They want a new birth. We have to keep our. And they have deep pockets. Their market cap is probably. I don't know what it is. 80, 90 billion. I don't know what it is.

Speaker B: But if they make the pie bigger

Speaker A: in the end, you know, in the end it's a net. Yeah. All of these things. Things. You know, when we saw Omnicool, which is the government sponsored research program. So Nvidia does all their two phase research under the guise of Cool Chip program or arpa. ARPA E program at University of UT Arlington or Villanova. There's a few big thermal kind of labs and they do. They're all in two phase on that. Verta, you know, showed this prototype and the conversation we had internally was maybe that's good, maybe that's a good thing. Two phase cdu. We don't, we don't know. I think it's a good thing.

Speaker D: Quick commentary on the air gap technology and you know, sovereign protection, which is a big thing now. And where do you see that market going?

Speaker A: I see that being the majority of the growth. I actually really wonder. And it doesn't really hurt us because we're not included in a lot of these right now. I don't know why we need 51 gigawatt data centers in the world. I don't know how LLMs achieve sustainable differentiation. And I ask the people about that. I ask that question a lot. I'm like, I'm confused. I'm kind of an infrastructure guy. Explain this to me. No one's been able to answer. So we will achieve HEI first. Okay, so someone else does it two months later. Why is that sustainable? Why, you know, are you going to be commodity? You're spending trillions, like \$1.5 trillion of investment on a product, on a chipset that's going to antique in three years. So your amortization schedule is way out of sync with reality, which maybe accountants should be, you know, weird.

Speaker F: That's the whole short argument for core weave and nebbious.

Speaker A: It's a big argument, but I think the bigger argument is what's the end goal? There's an arms race. Who are the winners and how do you sustainably win? You know, OpenAI was everything until three months ago. Now anthropic, it's all anyone talks about.

Speaker B: Ultimately, one AGI model cannot be that significantly better than the second one.

Speaker A: I think so. And if the second one is slightly better for certain things, then someone else will leapfrog. And so I read an article a couple days ago, maybe yesterday morning, it was really interesting and it exposed a lot of this. It said that they agree probably most LLMs were open source the next two to three years. Not a lot of margin in open source. The movement, the investment is going to be SLMs within companies and inference. Now all that plays to us, it actually plays to the world too because Again, how many 1 GW data centers? It was sort of a lot of chest thumping. Well, I'm going to build a 2 gigawatt data center. And then you had Saudi Arabia, we're going to build a 10, 10 gigawatt data. We're like, can we all just take a breath? Why are we doing this? And you invest, you know, you invest hundreds of billions into a gigawatt. When is your return on investment horizon? Your chips are going to be antiquated in three years. And by the way, you can't just stay with them forever because. Because the new chipset is 25 times better than that one. So I think Nvidia, this is purely me talking, what do I know? But I think Jensen, in his drive to dominate the world has forgotten some basic lessons in product life cycle, product launch. You know, intel knew how to pace that back in when intel was the great one, right? They knew that server OEMs could only absorb product design, product release, product, the client, the client base could only absorb product. The client base wanted some time with that product. They didn't want it antiquated in three months. So Nvidia is rolling out new chipset every year. Well, how do you, you have this sort of neurasthenia happening. You have literally people say I don't know what to buy, I don't know what to buy. JP Morgan announced that they have a moratorium on server purchases because they just don't know what to buy. They're just not doing anything. They're just not gonna buy. And also memory pricing and, you know, we have a GP server. We've offered \$400,000. The next version is going to be over \$900,000. So that math, how do you, how does that affect your ROI math? I think there's a, there's a lot of really abnormal market behavior in this and, you know, again, maybe there's something I'm missing. I don't know. I think the OpenAI IPO is going to expose a lot. I don't. I. My guess, and I'm probably, I'm usually wrong about these things. It's probably not going to be a very successful ipo. I think people are going to look at the filings and say, what on this? Math is not math. What is going on here? And then everyone's going to scrutinize everything else. The sector is going to take a breath. I think that'll be good for the world. I think it'll be great for us. But who knows?

Speaker B: You look at Claude's revenue round and it is like never seen before.

Speaker A: Spectacular, right?

Speaker B: Afford 30 billion in three months.

Speaker A: Yeah, it's incredible. It's spectacular. And look at parties for AI talking

Speaker B: about, that's, that's, that's the win case, in my opinion, for enterprise, you just don't need people. The ROI for companies.

Speaker A: Yeah, but then what happens? Then it gets in.

Speaker B: Yeah, it's a total thing.

Speaker A: But socialism, societal collapse. Have you had an influence? Inbound calls versus your sales calls? Yeah. So where are they? Where are they hearing about you? So we are, you know, if you, if you want to prepare for a tipping point, I think we're doing the right thing. So product line, everyone knows us that really, we are invited to the table already, which is pretty amazing for company. So our marketing team has done a wonderful job of getting our name out. We're at the right shows, we do a lot of earned media, you know, so we get a lot of exposure. I don't think anyone doesn't know Excelsius who's in this space at all, which is pretty amazing. So when a tipping point happens, we already have some brand equity, which I think is going to serve us really well. That's why, you might ask, why we're so focused on events and earned media and getting out there. It's because we want a great product line, great technology. We want a technology moat we want a great team and we want a great brand. And then when something big happens, we're just ready. We're ready for it. Right. That's how I'm thinking about it.

Speaker B: Is it, I don't want to put you too much on the spot here, but is it realistic to assume within the next 18 months that you'll get an order from a very recognizable name in market?

Speaker A: Yeah, yeah, absolutely.

Speaker C: Okay.

Speaker A: Absolutely. By the way, Dell's invited us in their executive briefing center. You'll see that released today or something.

Speaker G: Oh, that's great.

Speaker A: Small thing. It helps

Speaker C: as you think about your revenue grab. How much of that includes retrofitting old water systems to two phase or is that just all new construction?

Speaker A: All of it's being built.

Speaker F: Yeah.

Speaker C: Do you think there's a tipping point, you know, this is probably further down the road that people would go back to two phase or do you think once people have kind of gone that architecture they're not going to switch?

Speaker A: No, I think they will go, I think they will go to two phase. That's why I really emphasize the point that we're plug and play with, with

Speaker C: single phase and so like all your numbers, don't assume that kind of tipping point. That's you're just, you're just assuming basically new build.

Speaker A: Yeah.

Speaker C: So not the, not the old market which is actually a bigger market.

Speaker A: It's a much bigger market right now. We had a conversation yesterday about one Colo player, very traditional, about 80 data centers. They're all air cooled and developing a program for things retrofit not all of them but replicable program to retrofit them into liquid cooling could be really exciting for them because then their client, their choice of clients just increases dramatically because right now they can't handle these power dense.

Speaker C: Is ROIC pretty attractive for them? Is what is return on capital? Would that retrofit attractive for them?

Speaker A: It is. There's a couple levers for that. One air code. Data centers are really inefficient. So they can save money by doing this and then expand their clients they can serve. So their ROI could be really, really attractive by doing this. Even though retrofits are famously expensive. This is something you're going to take to them or they asked for. So we took it to them over the last two to three days to their CEO and he's excited about it.

Speaker D: How long is the conversation to go

Speaker A: to get to that point? Three days. So just a cold call? No, it was, no, no, that's not fair. We, we hired his ex Chief revenue officer and there happened to be a channel show this week that everyone was there and, and my CRO and, and her got in front of the CEO and said what, what if we do this, something like this? And he said, well, I was just thinking, I've been thinking about how do I, I'm stuck in this sort of old framework. How do I get this going? So I think timing was everything, but it wasn't a long, long cycle. Now before we

implement, before we develop, it's going to be a few months. What's the minimum cost for that guy to try it out? He wants to put in, put in a couple racks. What's it cost to try it out? Going to hook for if it fails? Oh, million million dollars to try it out and show his clients and make sure they're comfortable with them. About a million dollars.

Speaker C: Going back to some of your. You made some comments about the environmental concern and you know, we don't have enough power.

Speaker A: Like how does that impact like that

Speaker C: transition from, you know, water to two phase and you know, does it get mandated? Because so many states, you know, is that going to be a tail? Is there something more there? I'm just curious if you could expand on that.

Speaker A: So it's amazing how much detail. The Senate is into kuma now and I've been invited to speak to the Senate by Commerce actually. And they're obviously lobbying hard for this for like if we're going to build these, let's make sure every wat counts. Right. Let's not allow slop out there and, and Coons and a few other senators are really like into this. And so there will be some regulation and some direction from the government, which is good. But I think that this is another thing that you guys were saying, why it's intuitive that you would switch if you're building a hundred megawatt data center and you can save 5 megawatts if you can add 5% more GPUs because we've given them that thermal capacity in that same power envelope, then that's huge for them. Right. No one's really. We talk about OPEX savings. No one's talking about OPEX savings in this market. Sort of like you can save 35% operating expense. That's not. They don't care. What they do care is that translates into about 5 to 7% more GPUs into that data center. Yeah, that's massive. Yeah. Can you talk a little bit about your plan to ramp manufacturing? Yeah. So again, we have a facility, we're ramping we'll be able to ramp to kind of volume that sports in 18 months. We are having ongoing dialogues with a couple large contract manufacturers that are more than willing to take our volume if we need it. From the. So our plan at scale is to do about 25% of our own manufacturing and we call it almost MPI manufacturing. We think it's important that we manufacture our own stuff first. Go through a few sites. You probably want to do this before you hand it to a cm. So really get the process, get the bugs, get the instructions down, make sure the supply chain is secure and resilient and then hand drift. So I think you'll see that happening next year. But they're absolutely happy to do it. But most cms are pretty traditional. Like how many years can you promise us? And there's a penalty if you don't meet that. We're not in that position yet.

Speaker F: What are the most logical competitors that you face? Like if there's this lag time and hyperscalers or whomever evaluate other technologies.

Speaker B: Right.

Speaker F: You're not the only next gen possibility. Do you see anything out there yet? Or like what about the embedded cooling in. In the microchip directly.

Speaker A: So we, we are better for that. So if we go direct to silicon cooling, it won't really be directed Silicon Valley kind of a heat spreader on. So right now there's a silicon, the silicon. There's a thermal interface material, there's a heat spreader, another thermal interface material, then the cold pipe. That's a lot of thermal resistance, which is bad to get a thermal penalty for that. The idea is that eventually you'll get rid of that top 10 top heat spreader and directly cool an etched initial heat spreader which reduces your thermal resistance maybe by 2/3, which gives you maybe 5 to 7 degrees Celsius of performance improvement. So that's that. Would you rather have a refrigerant on your silicon or would you rather have water on your silicon? So I think that's a gift to us. There's a micro channeling like you do it within the silicon itself. You create channels. That's a gift to us too. I don't see any problem with that at all. That's a ways away because these fabs are incredibly complex and it's not going to happen next spring. I mean these things take a lot of planning. I think the first thing will be get rid of the bottom of the plate and have an etched heat spreader that you and you want to spray non conductive fluid on. That probably. Rather than conductive fluid, has there been

Speaker F: any movement on your board, or are

Speaker A: there any folks out there that you're in conversations with to add to the board or the team? So the newest board members, one CFO of LeGrand, because I got a board seat with investment, and then a guy who runs about half of Johnson Controls. He actually runs the other. Not the data center side. He runs fire suppression and controls and all that. Because they wanted that to be removed from the data center pipeline conversations, which I think legally is probably a smart move. We brought in Philip Morangela, who's one of the kind of founding management team of Edge Connect, which is a wholesale data center. He's added a huge amount of value over the next few months. I want to reconstitute the board a little bit. So replace a couple of the adventure guys. They'll still have board members. They'll still be legally their representative on the board. But we have enough of a name now that we can get some heavy hitters in the industry on that board. And that's. So by. Certainly by the end of this year, I want a reconstituted board. And the adventure guys, by the way, being perfectly gentleman who. That they're like, yeah, we don't. Is the former Vertiv sales guy still on your board? Yeah. Okay. He'll stay. Yeah. Yeah, he'll stay. He adds a lot of value as well. Yeah, they're really. I think it's Philip and John Hewitt is his name. Yeah, they. They are the best. I like a lively board. I mean, I. It makes me nervous if I say, here's what we're going to do, and everyone goes, that sounds like a good idea. That makes me

really nervous. So I like a lot of people with a lot of strong perspectives. And we're not quite there yet, but we'll get there.

Speaker F: I want to be conscious of time,

Speaker A: take some time off the tour, so we could definitely keep the questions going as we. Yeah. Was that helpful so far. There? I think it's 30 years, so kind of kick this off. Big important. We sublet this home Clorox. So they moved in during COVID and then moved out. Left all their furniture. We conveyed it all. They had a lot of silly slogans around. You know, teamwork has been working on our cinema engineers didn't really jive. They won the big trophy there. Over and back, we kept. It's a big plastic trophy. It has coffees for posers, which they award. I mean, you could. This is like a scene from the Office or Something but it's printed. So we kept it and it's actually become kind of. We award it every month to someone who's done something outstanding. And it's gone from kind of a joke to kind of just part of our culture. But that, that stays. But this is mostly mechanical engineering. Test engineering is there, finance is over there, firmware is over there, Double E is over there. And then Most of our PhDs around here do mostly thermal architecture. So you do mathematical simulations and so they should work and parts. We prototype it, we test it. They do a lot of. And then it goes over to the guys who. Mechanical engineers who package it and the firmware and electronic engineers who develop control boards, etc. Some tenants of our design are. We don't want to design each product as a standalone. We design platforms. So our controller boards are the same throughout our products. So we don't want a lot of tech debt. Right. One controller board extra I owe. So when we go bigger and bigger, we have that. Our firmware and software platforms throughout all of our products. Our screens, our GUI's are the same for all of our products. The standard protocols we use to connect to bms or data center management systems are all the same across all of our products. So you can plug any of those in to almost any dim system out there. So we're trying to use our age to. And lessons learned in the past to avoid some of that tech that. So this was our first product. Some of you have seen this before. It was designed for 100 kilowatt rack and could still support almost every torque load in the world. But no one wants to buy it because it's only 100 kilowatts. And so we've kind of discontinued this product. We're still using it. Inexpensive proof of concept. They want to play with it. So we just sent one to Paycom. For example, pacom is really interested in two face cooling. They didn't want to spend the money on a test for 250. We sent them a rack with one of these. And by the way, they're loving it. So that's. That's going to be a good account for us as well. This is actually the mezzanine under here. But it's. These are the caper things. These are actually CPUs. But you can see how everything works. The plate goes on that, the tubing goes into a manifold and then out the back comes those two vapor hoses and one fluid. So fluid, obviously you have to allow for vapor expansion. So that's

why we need Bigger hoses and the fluid goes in, goes here, vaporizer comes out, goes into the vapor manifold. Right. And we'll show you more back there. Basically that's this theory. And then if you. Lab looks like a label, but this we call these single loop testers. They're designed really to support. We test fluids. So SK gives us a fluid or Honeywell gives us a new fluid in each gen. We test the thermal properties of it. This is actually these plates here that we're testing are. Dell. Dell brought us single phase low plates. Is that still what's. Yes. I want to get this wrong. And this is what we're testing. So Dale wants us. So they're like, wow, that's kind of. They will bring you some plates. And so they brought us some plates. These are designed for water, not designed for us at all. For testing we'll compare the performance with water and then which we've already tested them with water based CPU as well. And then we test them with our fluid and our. Our technology and see which performs better if they perform the same. Great. So far we tend to perform better even on single phase goal plates. So. And then these single link testers are for the hyperscaler. So this is sure that you can't guess what kind of name that is. And you can see here. I don't know if you can. Sorry, OSHA would hate me for this. But this plate we designed about 17 plates for kiwi for their accelerators. We're just trying to optimize now. Then they'll go into a CDU test with this and then this one. I may be wrong. I think it's Meta. Or maybe we shipped. Did we ship meta? Ship Meta. So that's ours. Yes. Okay. I get strong every time. Thanks for helping. But Meta has. We actually designed one of these for Meta and they're using it in their lab now to test kind of two phase technology to cool their acceleration. Do they give you feedback down there? Oh yeah. The teams meet once a month. Yeah. Yeah. And so far they like it for Meta less fine for Apple quite a while. And you got kind of two teams involved with Apple. Some love us, some want to prove us wrong. I mean you just have to deal with that. Right? So we. But Meta seems to be kind of pretty excited about all this. I think Meta's going to move faster. I think in the end Microsoft's probably going to move fastest. Even though we're not tracked with Microsoft now and they keep having rifts that you know, will build a relationship with the team. And then they'll lay them all off. I mean, Microsoft's just doing it. It's like Dell just laying people off. Laying people off. Same. Same with aws. AWS we're engaged with now because of the grand. It's the power of that relationship. So if a global account manager on aws, he brought us in, the cooling team was based in Austin. We'd never met them. So it's power of these partnerships. Right.

Speaker F: But again, what's the alternative? Looking out two, three, four, five years. Like, my Apple guy wants to prove you wrong. Okay, what are they doing or where are they going?

Speaker A: But you have these. Really, what's the expression? Often wrong, but never in doubt. You have a lot.

Speaker F: Know it all, know nothing.

Speaker A: We call them the smartest guy in the room. We actually train our sales team to sell to the smartest guy in the room. And by that we mean the guy who's like, we don't need this. I never see a league. Well, so there's always one of those. Always the smartest guy in the room. What's the alternative? There's. There's not one over that time frame. In 10 years, you'll probably have completely photonic chips and things like that. Are you working towards milestones or how do you avoid getting mired down in

Speaker D: like an academic to academics conversation?

Speaker A: Have you been in this before? Because that's a great question. Because it tends to happen. So with. With Meta, we had that conversation. Like we're. We're doing all this testing design, like kind of should have get off the pot. And they seem to be getting off the pot. They're like, no, you're. You're right. Absolutely. We should buy your CDUs. We should have a bigger pocket. Apple were having that conversation with. We haven't quite gotten them off the clock, as it were, but it's a danger. It's an absolute danger. We want to be. Someone's in our shop. Exactly. It's a fear. And the fluid manufacturers, by the way, they all want to have us do their testing. I'm like. I'm like, guys, we could end up being an independent lab and not getting anything done. That's why we're going now. Like sk, we're like. Our business is selling. You bring this into SK Telecom for at least 10 megawatts of deployment and then we'll do this testing on your next map. So we're having very direct conversations to try to protect ourselves. Oh, if you want to see new plate. But I appreciate that. So this is so an Australian Company thinking we're not going to produce plates

Speaker E: in the long term.

Speaker A: We'll design them and then we'll license the technology probably for free because we want it to be adopted. But this is a plate that is printed. It's a 3D printed plate which means you can have a lot of very complex internal geometries. It's printed by a company in Australia. Another advantage of this there's some disadvantages because of aluminum Thermal resistance of aluminum is about twice copper but they've reduced the plate size because aluminum is stronger. So we reduce some of that. This would be in our load sleds for 2 megawatt load bank. But the nice thing is if we perfect this we don't have tariffs. They'll actually put a couple of their printers in our factory and they'll load the files in Australia and print them here. We avoid 55% tariffs. We pay. We pay Taiwan on grace plates right now. 55% tariffs. I don't know how any hard tech company this company with all these tariffs.

Speaker B: Who makes those printers?

Speaker A: Sorry, who makes those printers? Chinese company. It's not one of the name brands that you've probably heard of but it's and copper powder is not ready for prime time

yet. It's too the porosity is the problem. So yeah from the ability of the refrigerant molecules but aluminum is pretty cool. But isn't it that's also super light. And one of Nvidia's concerns is Iraq now weighs about 5,000 because of all the copper plating. So and that's an issue. Right. So aluminum is much much lighter. So we're trying to stay ahead of the curve on experimentation again partner with two partners as well. So let's go this way.

Speaker F: The scene now is actually pretty good. The Italian all French pastas some more boxes When I moved there it was absolutely awful really.

Speaker A: So it's grown quite a lot since

Speaker F: the whole my family Houston⁸⁹ that was similarly awful. And now.

Speaker H: Even though it's in your. I'm not saying here I'm saying from

Speaker B: a customer perspective the customers want to

Speaker H: Dell so we have to work that. But that's one of the. Integrate in their factory. That's where we want to get a lot of space right now. So it starts connect or have to go just take the risk themselves. But we hope that dark will have a volume that we're getting. So we hope everyone lined up on. \$25 billion. And that gets rid of all that's got an inflection point when Nvidia goes

Speaker B: your valuation.

Speaker H: Yeah.

Speaker A: So this, You can slide the zip

Speaker H: box out, stand there comfortably and place modules on it. The pump is also. You isolate through that. And then these are all sleds. So you can sit in the aisle, place the pump comfortably. We try as much as possible off the shelf. But this reservoir, because of the packaging.

Speaker A: Packaging is custom. The hosing is cut to a custom length. And the plan,

Speaker F: there's some good people.

Speaker H: You just got to weed through them. Like 95% of this is scanner, which means we can drive that.

Speaker A: Mike.

Speaker H: Oh, mike.

Speaker A: Here, keep it.

